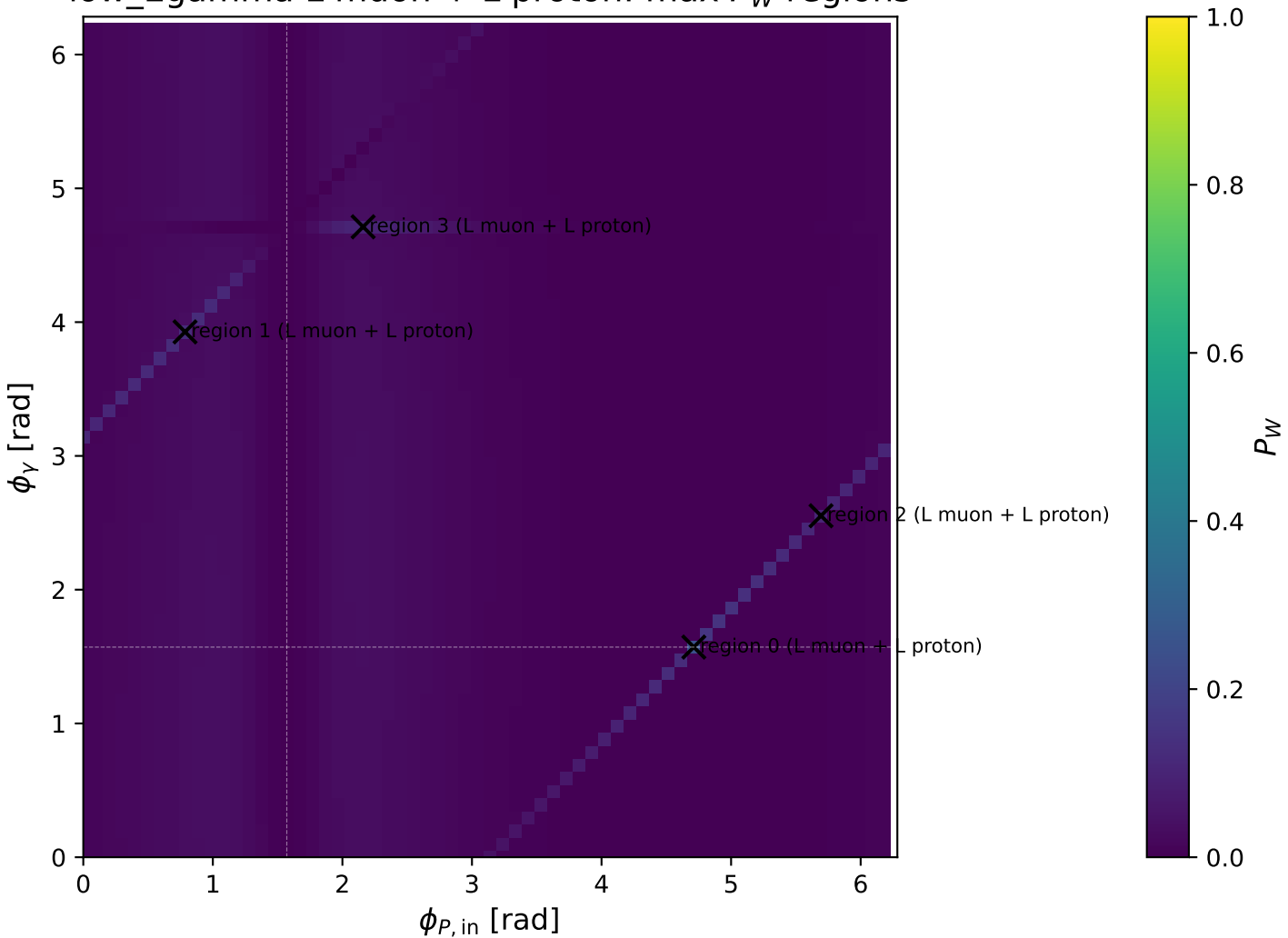
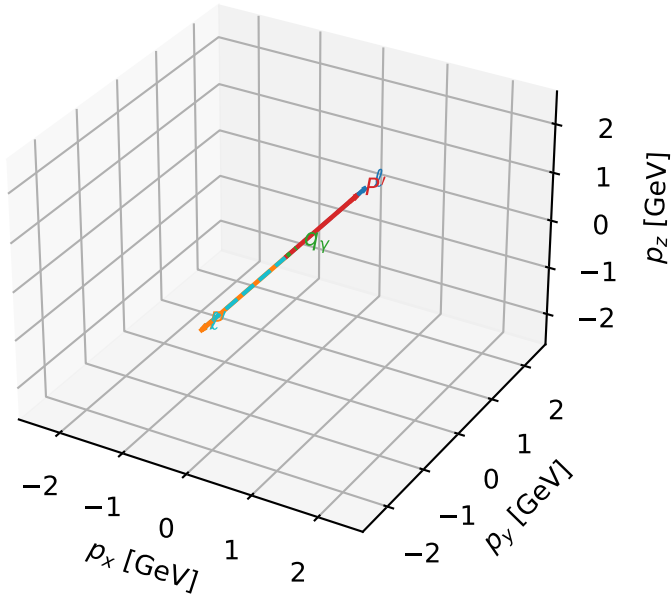


low\_Egamma L muon + L proton: max  $P_W$  regions



# L muon + L proton characteristic max $P_W$ configuration

3D momenta

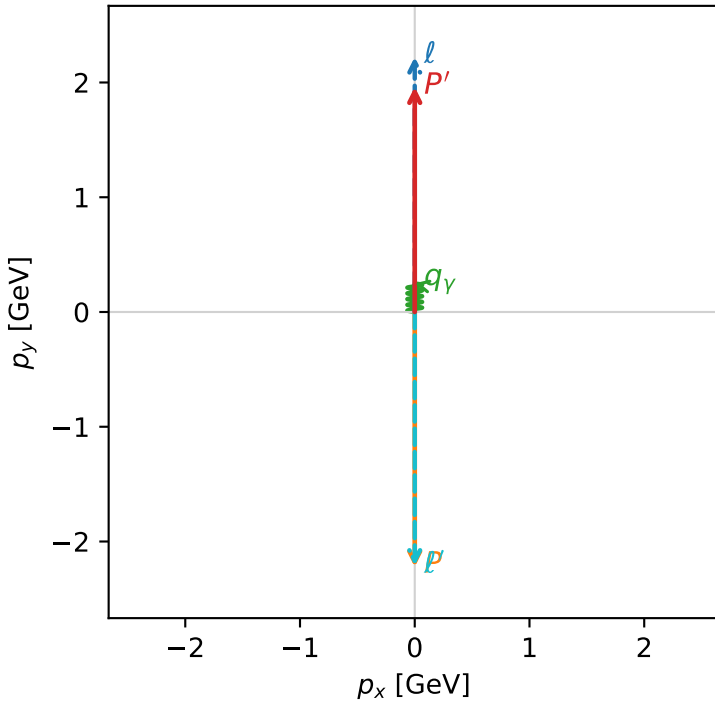


w\_purity\_low\_egamma\_ll\_region\_0 P\_W=0.248262  
 region: low\_Egamma\_LL\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

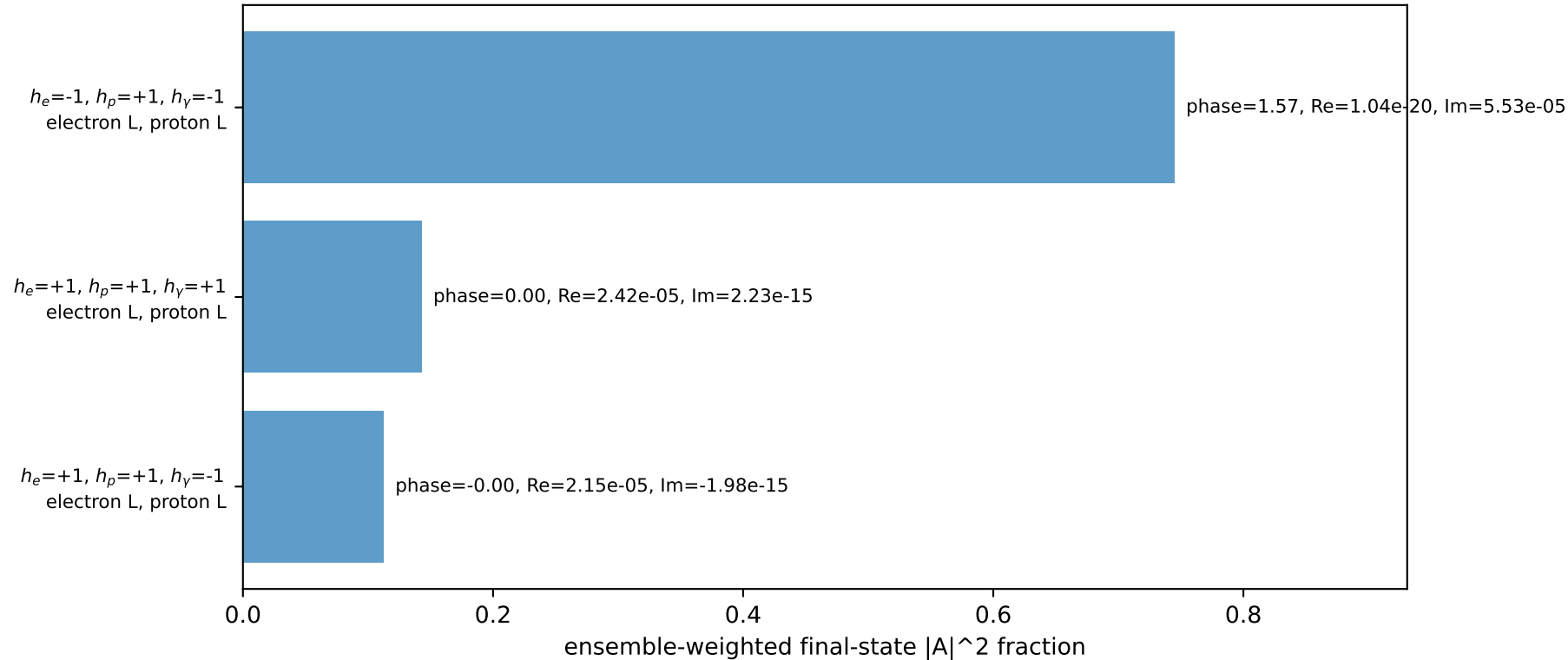
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=1.96163$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=1.5708$   
 $Q^2=19.6668$ ,  $x_B=0.994621$ ,  $t=-17.4552$ ,  $W^2=0.986208$ ,  $y=0.958708$   
 $\theta(\ell', \gamma)=180$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2256$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2142$   $p_x= -0.0000$   $p_y= -2.2116$   $p_z= 0.0000$   
 $P'$   $E= 2.1744$   $p_x= 0.0000$   $p_y= 1.9616$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0000$   $p_y= 0.2500$   $p_z= 0.0000$

Transverse plane

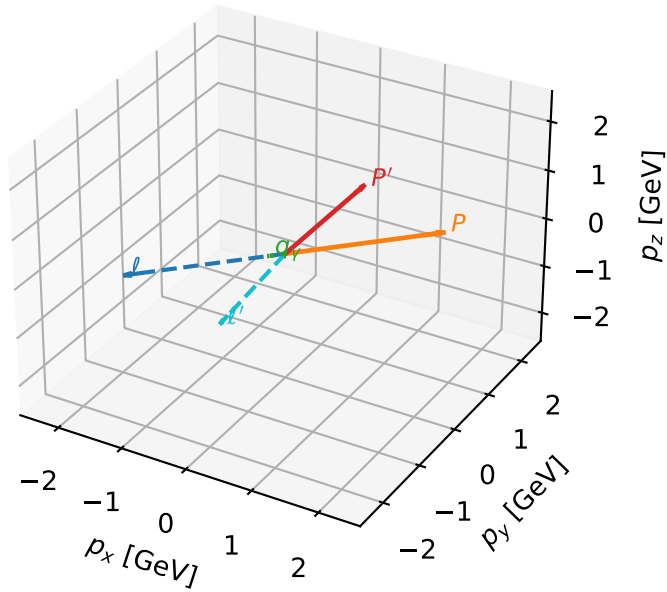


Leading initial-ensemble/final-state components (N=3, retained=100.0%)



# L muon + L proton characteristic max $P_W$ configuration

3D momenta

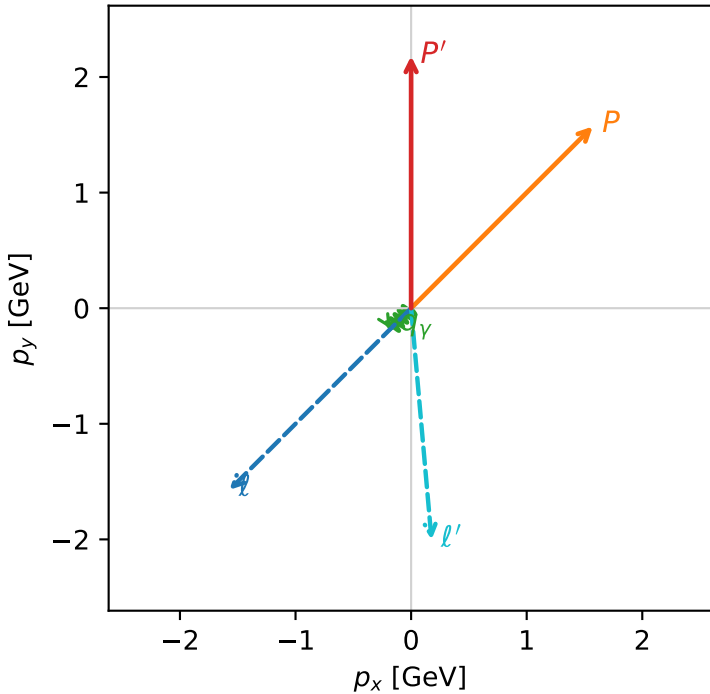


w\_purity\_low\_egamma\_ll\_region\_1 P\_W=0.123387  
 region: low\_Egamma\_LL\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

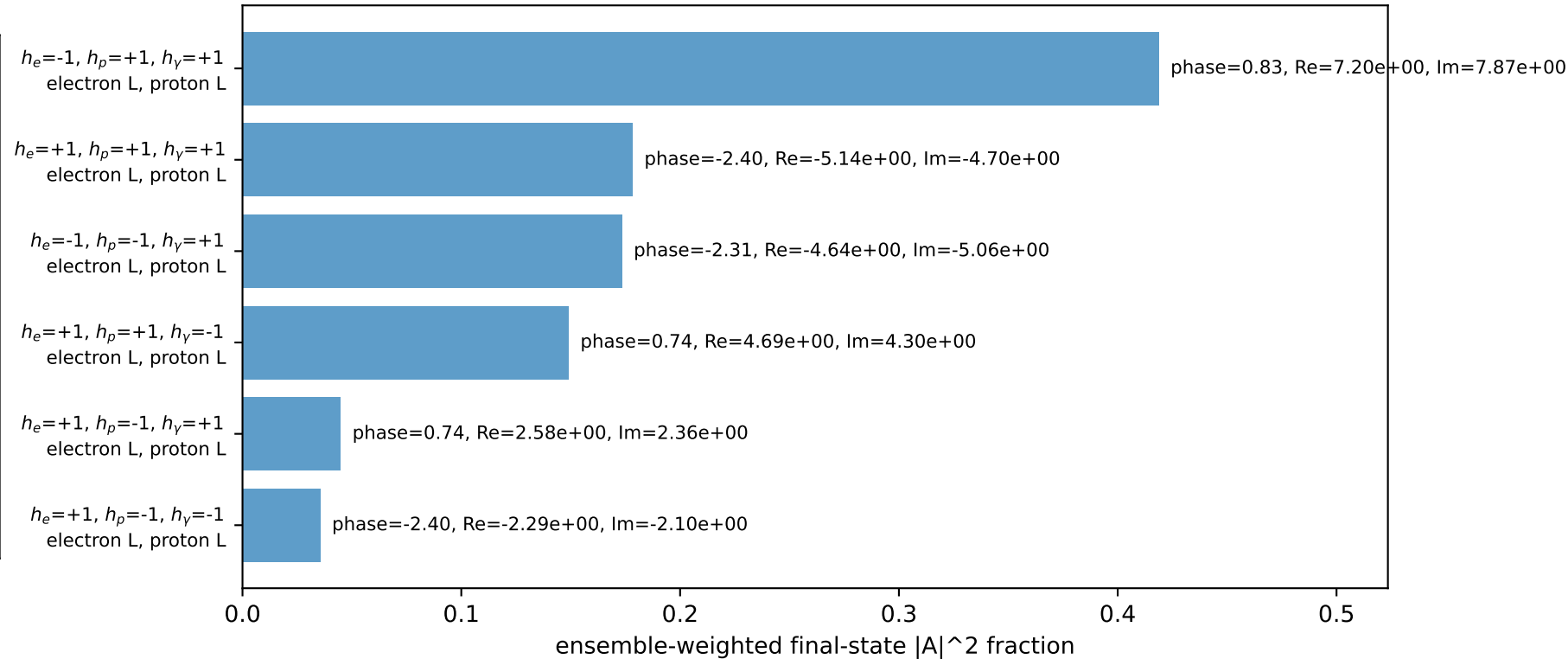
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.18078$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.92699$ ,  $\phi_P=0.785398$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=3.92699$   
 $Q^2=3.20024$ ,  $x_B=0.620413$ ,  $t=-2.84024$ ,  $W^2=2.83785$ ,  $y=0.250099$   
 $\theta(l', \gamma)=50.0411$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -1.5720$   $p_y= -1.5720$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 1.5720$   $p_y= 1.5720$   $p_z= 0.0000$   
 $l'$   $E= 2.0146$   $p_x= 0.1768$   $p_y= -2.0040$   $p_z= 0.0000$   
 $P'$   $E= 2.3740$   $p_x= 0.0000$   $p_y= 2.1808$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.1768$   $p_y= -0.1768$   $p_z= 0.0000$

Transverse plane

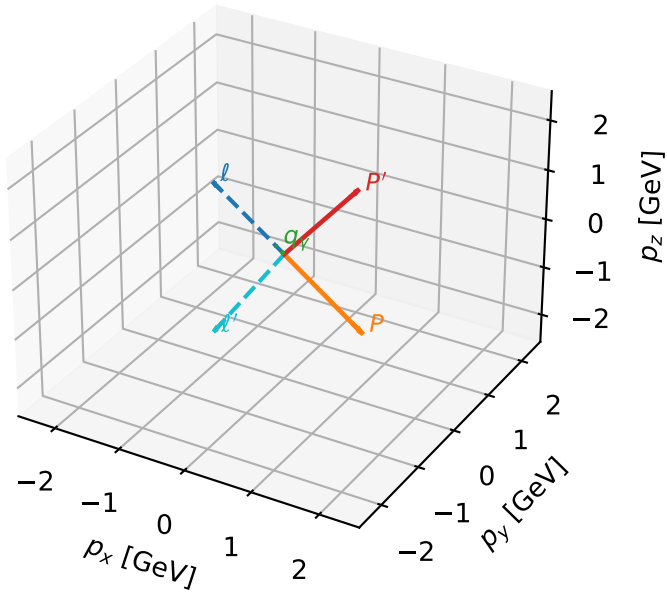


Leading initial-ensemble/final-state components (N=6, retained=100.0%)



# L muon + L proton characteristic max $P_W$ configuration

3D momenta



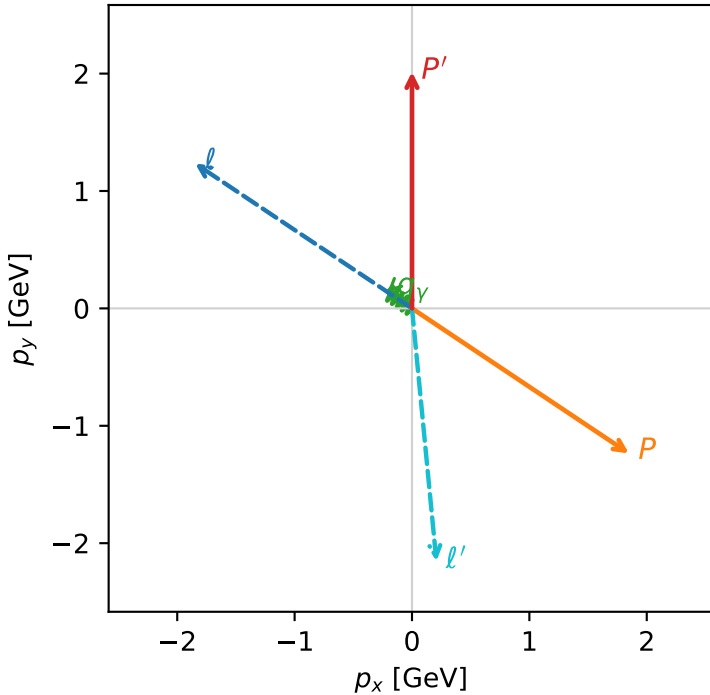
w\_purity\_low\_egamma\_ll\_region\_2 P\_W=0.105675  
 region: low\_Egamma\_LL\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.01469$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.55254$ ,  $\phi_P=5.69414$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=2.55254$   
 $Q^2=15.708$ ,  $x_B=0.96608$ ,  $t=-13.9416$ ,  $W^2=1.43137$ ,  $y=0.788349$   
 $\theta(l', \gamma)=129.263$  deg

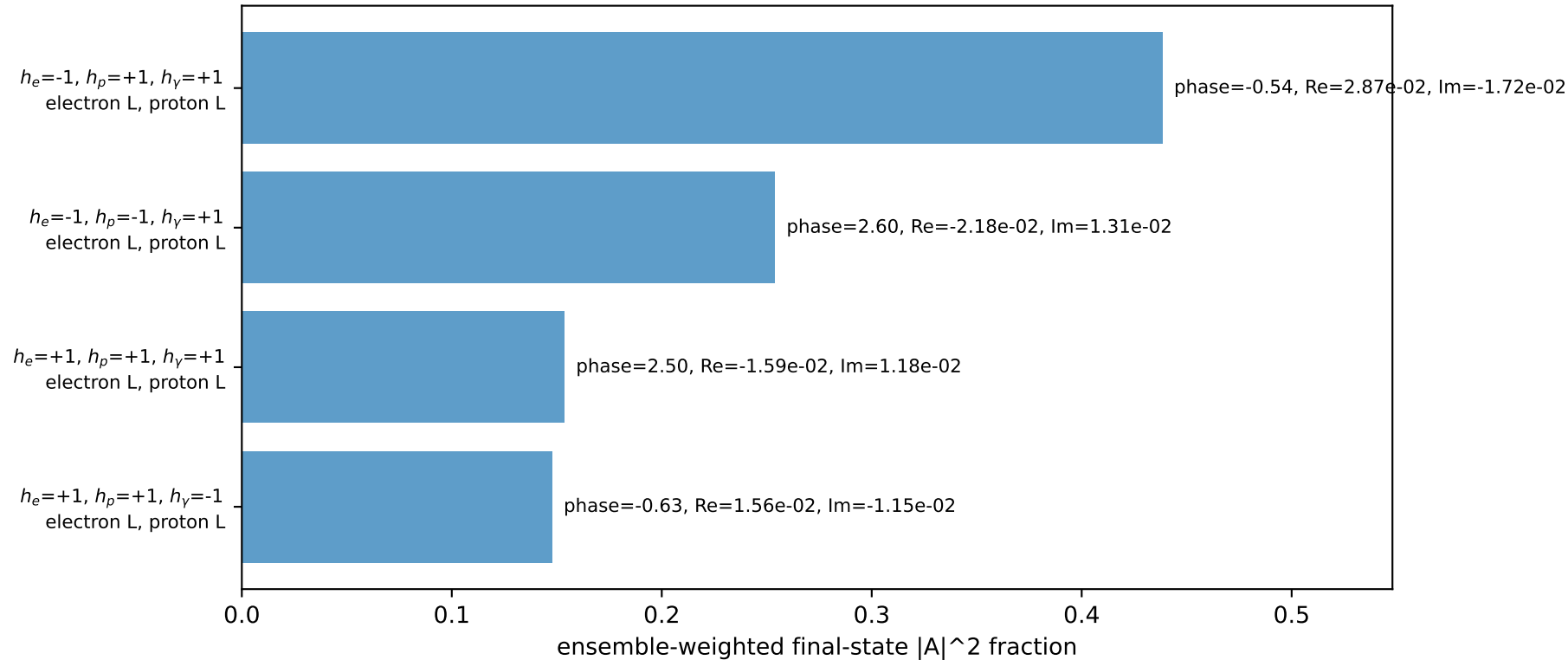
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.2256$   $p_x= -1.8484$   $p_y= 1.2351$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 1.8484$   $p_y= -1.2351$   $p_z= 0.0000$   
 $l'$   $E= 2.1662$   $p_x= 0.2079$   $p_y= -2.1536$   $p_z= 0.0000$   
 $P'$   $E= 2.2223$   $p_x= 0.0000$   $p_y= 2.0147$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2079$   $p_y= 0.1389$   $p_z= 0.0000$

Transverse plane

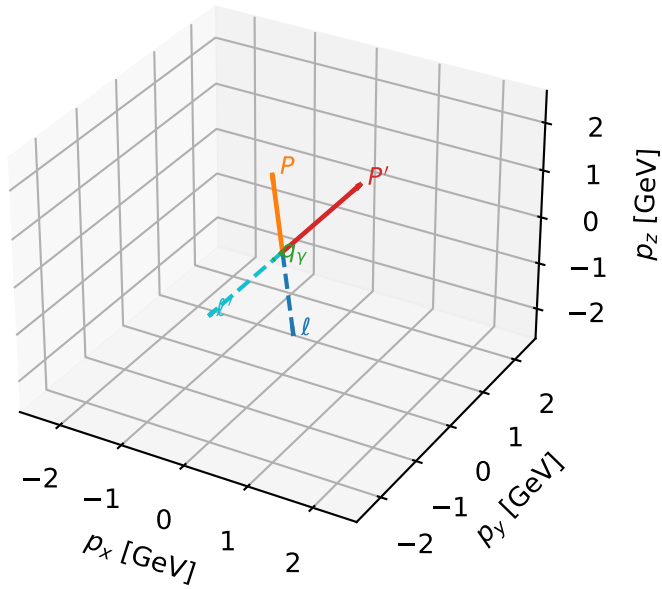


Leading initial-ensemble/final-state components (N=4, retained=99.4%)



# L muon + L proton characteristic max $P_W$ configuration

3D momenta

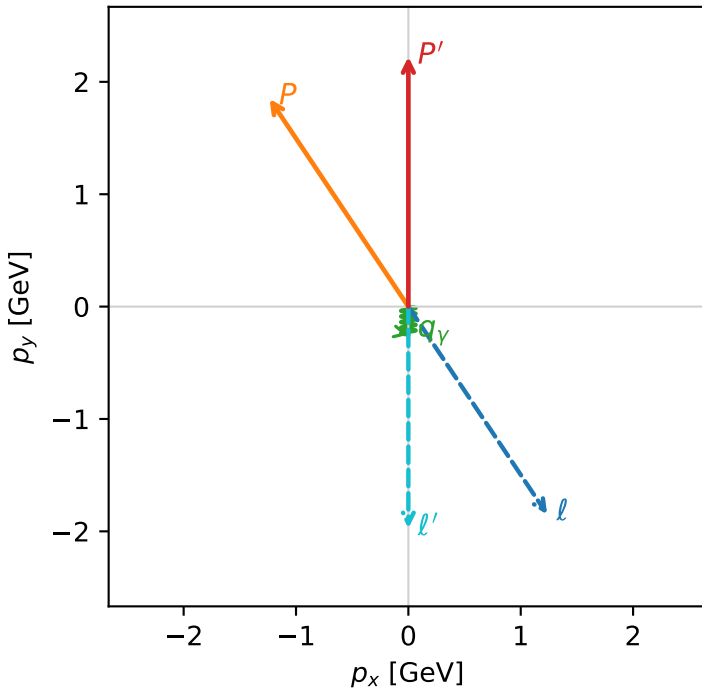


w\_purity\_low\_egamma\_ll\_region\_3 P\_W=0.103737  
region: low\_Egamma\_LL\_region\_3  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.22295$   
 $\theta_{in}=1.57079$ ,  $\phi_l=5.30144$ ,  $\phi_p=2.15984$   
 $E_Y=0.25$ ,  $\phi_Y=4.71239$   
 $Q^2=1.47853$ ,  $x_B=0.389459$ ,  $t=-1.66571$ ,  $W^2=3.19769$ ,  $y=0.184068$   
 $\theta(l', \gamma)=0$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 1.2351$   $p_y= -1.8484$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -1.2351$   $p_y= 1.8484$   $p_z= 0.0000$   
 $l'$   $E= 1.9758$   $p_x= 0.0000$   $p_y= -1.9729$   $p_z= 0.0000$   
 $P'$   $E= 2.4127$   $p_x= 0.0000$   $p_y= 2.2229$   $p_z= 0.0000$   
 $q_Y$   $E= 0.2500$   $p_x= -0.0000$   $p_y= -0.2500$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=100.0%)

